

Puppet Mastery Part-9:

Extensive practice-1: Production-Grade Puppet Infrastructure Orchestration: Dynamic Multi-Environment Configuration, Fact-Driven Automation & HAProxy Load Balancing

Master:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/loadbalancer/manifests$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.8.85)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-master
Info: Applying configuration version '1787124122'
Notice: /Stage[main]/Loadbalancer::Report/File[/var/lib/loadbalancer]/ensure: created
Notice: /Stage[main]/Loadbalancer::Report/File[/var/lib/loadbalancer/loadbalancer-report.txt]/ensure: defined content as '{sha256}f83567319f7f268052d908ad6c6e16ed03ceff273bf986f8acc8588e45d59a72'
Notice: Applied catalog in 0.55 seconds
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/loadbalancer/manifests$ sudo systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/lib/systemd/system/haproxy.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-08-19 07:19:16 UTC; 3min 21s ago
     Docs: man:haproxy(1)
           file:/usr/share/doc/haproxy/configuration.txt.gz
   Process: 2023 ExecStartPre=/usr/sbin/haproxy -Ws -f $CONFIG -c -q $EXTRA_OPTS
   Main PID: 2026 (haproxy)
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
  
<p>Managed automatically by Puppet.</p>  
</body>  
</html>  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba  
lancer/manifests$ for i in {1..15}; do curl -s http://localhost | grep '<h1>';do  
ne  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
    <h1>WEB-01</h1>  
    <h1>WEB-02</h1>  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba  
lancer/manifests$ |
```

Note: Successful load balancing between the two agents

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ sud  
o cat site.pp  
node 'puppet-master'{  
  include loadbalancer  
}  
node 'puppet-agent01'{  
  include dynamicinfra  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/manifests$ cd  
..  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules/lo  
adbaler  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba  
lancer$ ls  
manifests  templates  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba  
lancer$ cd manifests  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba  
lancer/manifests$ cat init.pp  
class loadbalancer(  
  String $environment,  
  String $server_role,  
  Hash $backends,  
  Integer $frontend_port,
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
String $config_path,
String $service_name,
String $report_path
) {

  class { 'loadbalancer::install': }

  class { 'loadbalancer::config':
    environment => $environment,
    server_role => $server_role,
    backends    => $backends,
    frontend_port => $frontend_port,
    config_path => $config_path,
  }

  class { 'loadbalancer::service':
    service_name => $service_name,
  }

  class { 'loadbalancer::report':
    environment => $environment,
    server_role => $server_role,
    report_path => $report_path
  }
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data

Class['loadbalancer::install']
  -> Class['loadbalancer::config']
  ~> Class['loadbalancer::service']
  -> Class['loadbalancer::report']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/loadbalancer/manifests$ cat config.pp
class loadbalancer::config(
  String $environment,
  String $server_role,
  Hash $backends,
  Integer $frontend_port,
  String $config_path
) {

  file { $config_path:
    ensure => file,
    content => epp('loadbalancer/haproxy.cfg.epp', {
      'environment' => $environment,
      'server_role' => $server_role,
      'backends' => $backends,
      'frontend_port' => $frontend_port,
    }),
  }
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
owner  => 'root',
group  => 'root',
mode   => '0644',
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/manifests$ cat install.pp
class loadbalancer::install{

  package{'haproxy':
    ensure=>installed
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/manifests$ cat service.pp
class loadbalancer::service(
  String $service_name
){

  service{$service_name:
    ensure=>running,
    enable=>true
  }
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/manifests$ cat service.pp
class loadbalancer::service(
  String $service_name
){

  service{$service_name:
    ensure=>running,
    enable=>true
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/templates$ ls
haproxy.cfg.epp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/templates$ sudo cat haproxy.cfg.epp
global
  log /dev/log local0
  log /dev/log local1 notice
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
log global
mode http
option httplog
option dontlognull
timeout connect 5s
timeout client 30s
timeout server 30s

frontend http_front
  bind *:<%= $frontend_port %>
  default_backend web_servers

backend web_servers
  balance roundrobin
  option httpchk GET /

<% $backends.each |$name,$backend| { -%>
  server <%= $name %> <%= $backend['ip'] %>:<%= $backend['port'] %> check
<% } -%>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd dy

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
functions manifests templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra$ cd manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ sudo cat init.pp
class dynamicinfra(
  String $server_role,
  String $environment_name,
  String $node_label,
  String $output_directory,
  Array[String] $packages,
  String $apache_config_path
){

  class{'dynamicinfra::install':
    packages=>$packages
  }

  class{'dynamicinfra::config':
    server_role=>$server_role,
    environment_name=>$environment_name,
    node_label=>$node_label,
    output_directory=>$output_directory,
    apache_config_path=>$apache_config_path
  }
}

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
}

class{'dynamicinfra::service': }

class{'dynamicinfra::report':
  server_role=>$server_role,
  environment_name=>$environment_name,
  node_label=>$node_label,
  output_directory=>$output_directory
}

Class['dynamicinfra::install']->Class['dynamicinfra::config']->
Class['dynamicinfra::service']->Class['dynamicinfra::report']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynami
infra/manifests$ sudo cat config.pp
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $node_label,
  String $output_directory,
  String $apache_config_path
){
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data

$hostname = $facts['networking']['hostname']
$ip = $facts['networking']['ip']
$os_name = $facts['os']['name']
$os_family = $facts['os']['family']

$normalized_role = dynamicinfra::normalize_role($server_role)

$system_data = dynamicinfra::prepare_system_data({
  'hostname'=>$hostname,
  'ip'=>$ip,
  'os_name'=>$os_name,
  'os_family'=>$os_family
})

$config_data = dynamicinfra::build_node_config($normalized_role,$environment_na
me,
  $node_label,$system_data)

file { $output_directory:
  ensure => directory,
  owner  => 'root',
  group  => 'root',
  mode   => '0755',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
file { "${apache_config_path}/dynamicinfra.conf":
  ensure => file,
  owner  => 'root',
  group  => 'root',
  mode   => '0644',
  content => epp('dynamicinfra/config.epp', {
    'config_data' => $config_data,
  })
}

file { '/var/www/html/index.html':
  ensure => file,
  owner  => 'root',
  group  => 'root',
  mode   => '0644',
  content => epp('dynamicinfra/web.epp', {
    'config_data' => $config_data,
  }),
  notify => Service['apache2']
}

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
infra/manifests$ sudo cat install.pp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
infra/manifests$ sudo cat install.pp
class dynamicinfra::install(
  Array[String] $packages){

  package{$packages:
    ensure=>installed
  }
}

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
infra/manifests$ sudo cat service.pp
class dynamicinfra::service{

  service{'apache2':
    ensure=>running,
    enable=>true
  }
}

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
infra/manifests$ sudo cat report.pp
class dynamicinfra::report(
  String $server_role,
  String $environment_name,
  String $node_label,
  String $output_directory
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
file { "${output_directory}/system-report.txt":
  ensure => file,
  owner  => 'root',
  group  => 'root',
  mode   => '0644',
  content => "Node: ${node_label}\nRole: ${server_role}\nEnvironment: ${environment_name}\n",
}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/manifests$ cd ..
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra$ cd functions
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/functions$ ls
build_node_config.pp      prepare_system_data.epp.save
build_node_config.pp.save prepare_system_data.pp
normalize_role.pp         prepare_system_data.pp.save
practicedsl.pp
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/functions$ cat build_node_config.pp
function dynamicinfra::build_node_config(
  String $server_role,

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
$role=dynamicinfra::normalize_role($server_role)

$system=dynamicinfra::prepare_system_data($system_data)

$config = {
  'server_role' => $role,
  'environment' => $environment,
  'node_label' => $node_label,
  'system' => {
    'hostname' => $system['hostname'],
    'ip'=>$system['ip'],
    'os'=>$system['os_name'],
    'os_family'=>$system['os_family']
  },
  'web' => {
    'root'=>'/var/www/html',
    'port' => 80
  }
}
$config
}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/functions$ cat normalize_role.pp
function dynamicinfra::normalize_role(String $role,String $default='web')>>

```



```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
String{
  $normalized_role=$role.downcase.strip
  if $normalized_role == "" { return $default }
  if $normalized_role in ['web','web_server','frontend'] {
    return 'web'
  } elsif $normalized_role in ['api','api_server','backend'] {
    return 'api'
  } elsif $normalized_role == 'loadbalancer' {
    return 'loadbalancer'
  } else {
    fail("Unsupported server role: ${role}")
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynamicinfra/functions$ cat prepare_system_data.pp

function dynamicinfra::prepare_system_data(
  Hash[String, String] $facts_data
) >> Hash[String, String] {
  {

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
  if $normalized_role in ['web','web_server','frontend'] {
    return 'web'
  } elsif $normalized_role in ['api','api_server','backend'] {
    return 'api'
  } elsif $normalized_role == 'loadbalancer' {
    return 'loadbalancer'
  } else {
    fail("Unsupported server role: ${role}")
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynamicinfra/functions$ cat prepare_system_data.pp

function dynamicinfra::prepare_system_data(
  Hash[String, String] $facts_data
) >> Hash[String, String] {
  {
    'hostname' => $facts_data['hostname'].strip,
    'ip'       => $facts_data['ip'].strip,
    'os_name'  => $facts_data['os_name'].strip,
    'os_family' => $facts_data['os_family'].strip,
  }
}

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
cinfra/templates$ cat config.epp
<VirtualHost *:80>
  ServerName <%= $config_data['node_label'] %>

  DocumentRoot <%= $config_data['web']['root'] %>

  <Directory <%= $config_data['web']['root'] %>>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
  </Directory>

  ErrorLog ${APACHE_LOG_DIR}/puppet-managed-error.log
  CustomLog ${APACHE_LOG_DIR}/puppet-managed-access.log combined

  Header always set X-Puppet-Environment "<%= $config_data['environment'] %>"
  Header always set X-Puppet-Role "<%= $config_data['server_role'] %>"
</VirtualHost>
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ cat web.epp
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
cinfra/templates$ cat web.epp
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title><%= $config_data['node_label'] %></title>
</head>

<body>
  <h1><%= $config_data['node_label'] %></h1>

  <h2>Puppet Managed Web Server</h2>

  <p><strong>Environment:</strong> <%= $config_data['environment'] %></p>
  <p><strong>Server Role:</strong> <%= $config_data['server_role'] %></p>

  <h3>System Information</h3>

  <p><strong>Hostname:</strong> <%= $config_data['system']['hostname'] %></p>
  <p><strong>IP Address:</strong> <%= $config_data['system']['ip'] %></p>
  <p><strong>Operating System:</strong> <%= $config_data['system']['os'] %></p>
  <p><strong>OS Family:</strong> <%= $config_data['system']['os_family'] %></p>

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
<h3>System Information</h3>

<p><strong>Hostname:</strong> <%= $config_data['system']['hostname'] %></p>
<p><strong>IP Address:</strong> <%= $config_data['system']['ip'] %></p>
<p><strong>Operating System:</strong> <%= $config_data['system']['os'] %></p>
>
<p><strong>OS Family:</strong> <%= $config_data['system']['os_family'] %></p>
>

<hr>

<p>Managed automatically by Puppet.</p>
</body>
</html>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ ls ..
functions manifests templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd data
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ |
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data
GNU nano 6.2 production.yaml
loadbalancer::environment: 'production'
loadbalancer::frontend_port: 80
loadbalancer::service_name: 'haproxy'
loadbalancer::config_path: '/etc/haproxy/haproxy.cfg'
dynamicinfra::environment_name: 'production'
dynamicinfra::output_directory: '/var/lib/dynamicinfra'
dynamicinfra::packages:
- apache2
dynamicinfra::apache_config_path: '/etc/apache2/sites-available'
loadbalancer::report_path: '/var/lib/loadbalancer/loadbalancer-report.txt'
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-agent01.yaml
dynamicinfra::server_role: 'web_server'
dynamicinfra::node_label: 'WEB-01'
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-master.yaml
loadbalancer::server_role: 'loadbalancer'

loadbalancer::backends:
  web01:
    ip: '172.31.5.133'
    port: 80
  web02:
    ip: '172.31.10.53'
    port: 80
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development/d...
GNU nano 6.2 development.yaml
dynamicinfra::environment_name: 'development'

dynamicinfra::output_directory: '/var/lib/dynamicinfra'

dynamicinfra::packages:
  - apache2

dynamicinfra::apache_config_path: '/etc/apache2/sites-available'
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development/d...
GNU nano 6.2 puppet-agent02.yaml
dynamicinfra::server_role: 'web_server'
dynamicinfra::node_label: 'WEB-02'
```

Puppet-Agent01:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.8.85)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787123221'  
Notice: /Stage[main]/Dynamicinfra::Install/Package[apache2]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/lib/dynamicinfra]/ensure: cr  
eated  
Notice: /Stage[main]/Dynamicinfra::Config/File[/etc/apache2/sites-available/dyna  
micinfra.conf]/ensure: defined content as '{sha256}610750acf1f344aced4783bdd304b  
42917769c98dadf65edeaad388961121b09'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content  
:  
--- /var/www/html/index.html      2026-08-19 07:07:08.932377217 +0000  
+++ /tmp/puppet-file20260819-1227-p18c75      2026-08-19 07:07:16.540455333 +0  
000  
@@ -1,363 +1,28 @@
```

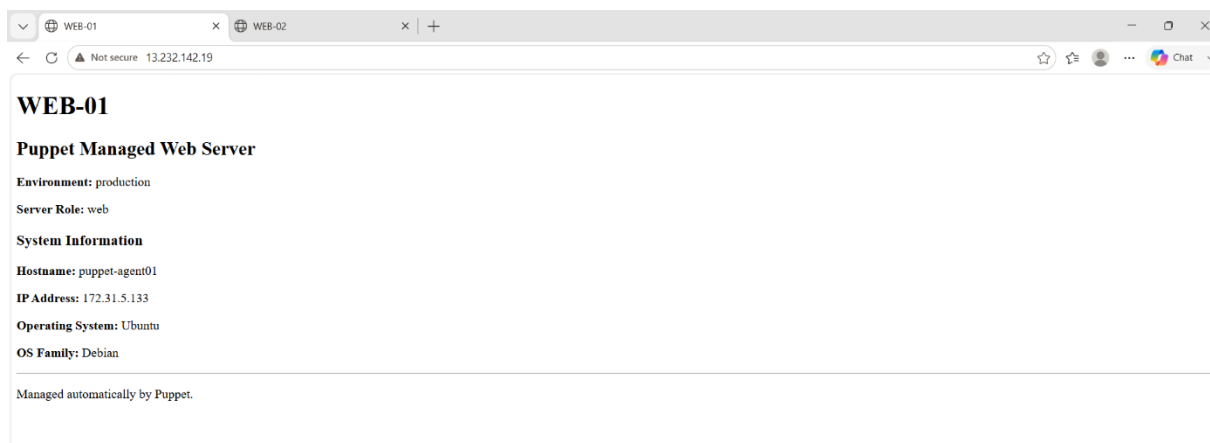
```
ubuntu@puppet-agent01: ~  
d8b5f7001'  
Info: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]: Scheduli  
ng refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'  
from 1 event  
Notice: /Stage[main]/Dynamicinfra::Report/File[/var/lib/dynamicinfra/system-repo  
rt.txt]/ensure: defined content as '{sha256}7c79b42877ebe52ce9a52ae9393f5c1595b9  
1ec5f25e091176ca69e6303b5773'  
Notice: Applied catalog in 12.70 seconds  
ubuntu@puppet-agent01:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese  
   Active: active (running) since Wed 2026-08-19 07:07:16 UTC; 16min ago  
     Docs: https://httpd.apache.org/docs/2.4/  
   Process: 2133 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU  
 Main PID: 2138 (apache2)  
    Tasks: 55 (limit: 1018)  
   Memory: 5.1M  
      CPU: 77ms  
   CGroup: /system.slice/apache2.service  
           └─2138 /usr/sbin/apache2 -k start  
             └─2139 /usr/sbin/apache2 -k start  
               └─2140 /usr/sbin/apache2 -k start
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ curl http://localhost  
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>WEB-01</title>  
</head>  
  
<body>  
  <h1>WEB-01</h1>  
  
  <h2>Puppet Managed Web Server</h2>  
  
  <p><strong>Environment:</strong> production</p>  
  <p><strong>Server Role:</strong> web</p>  
  
  <h3>System Information</h3>  
  
  <p><strong>Hostname:</strong> puppet-agent01</p>  
  <p><strong>IP Address:</strong> 172.31.5.133</p>  
  <p><strong>Operating System:</strong> Ubuntu</p>  
  <p><strong>OS Family:</strong> Debian</p>
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo ss -tlnp | grep :80  
LISTEN 0      511          *:80          *:~      users:(("apache2",pid=2140,fd=4),("apache2",pid=2139,fd=4),("apache2",pid=2138,fd=4))  
ubuntu@puppet-agent01:~$ hostname -i  
172.31.5.133 fe80::8c7:7eff:fe34:9163  
ubuntu@puppet-agent01:~$ ^C  
ubuntu@puppet-agent01:~$ sudo ls -l /var/lib/dynamicinfra  
total 4  
-rw-r--r-- 1 root root 54 Aug 19 07:07 system-report.txt  
ubuntu@puppet-agent01:~$ sudo ls -l /etc/apache2/sites-available/dynamicinfra.conf  
-rw-r--r-- 1 root root 453 Aug 19 07:07 /etc/apache2/sites-available/dynamicinfra.conf  
ubuntu@puppet-agent01:~$ sudo ls -l /var/www/html/index.html  
-rw-r--r-- 1 root root 652 Aug 19 07:07 /var/www/html/index.html  
ubuntu@puppet-agent01:~$ sudo systemctl status apache2 --no-pager  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)  
   Active: active (running) since Wed 2026-08-19 07:07:16 UTC; 26min ago  
     Docs: https://httpd.apache.org/docs/2.4/  
   Process: 2133 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)  
   Main PID: 2138 (apache2)
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo cat /var/lib/dynamicinfra/system-report.txt  
Node: WEB-01  
Role: web_server  
Environment: production  
ubuntu@puppet-agent01:~$ sudo cat /etc/apache2/site-available/dynamicinfra.conf  
cat: /etc/apache2/site-available/dynamicinfra.conf: No such file or directory  
ubuntu@puppet-agent01:~$ sudo cat /etc/apache2/sites-available/dynamicinfra.conf  
<VirtualHost *:80>  
    ServerName WEB-01  
  
    DocumentRoot /var/www/html  
  
    <Directory /var/www/html>  
        Options Indexes FollowSymLinks  
        AllowOverride None  
        Require all granted  
    </Directory>  
  
    ErrorLog ${APACHE_LOG_DIR}/puppet-managed-error.log  
    CustomLog ${APACHE_LOG_DIR}/puppet-managed-access.log combined  
  
    Header always set X-Puppet-Environment "production"  
    Header always set X-Puppet-Role "web"  
</VirtualHost>
```

```
ubuntu@puppet-agent01: ~  
</head>  
<body>  
    <h1>WEB-01</h1>  
  
    <h2>Puppet Managed Web Server</h2>  
  
    <p><strong>Environment:</strong> production</p>  
    <p><strong>Server Role:</strong> web</p>  
  
    <h3>System Information</h3>  
  
    <p><strong>Hostname:</strong> puppet-agent01</p>  
    <p><strong>IP Address:</strong> 172.31.5.133</p>  
    <p><strong>Operating System:</strong> Ubuntu</p>  
    <p><strong>OS Family:</strong> Debian</p>  
  
    <hr>  
  
    <p>Managed automatically by Puppet.</p>  
</body>  
</html>
```



Puppet-Agent02:


```

ubuntu@puppet-agent02: ~
ubuntu@puppet-agent02:~$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Refreshing CA certificate
Info: CA certificate is unmodified, using existing CA certificate
Info: Refreshing CRL
Info: CRL is unmodified, using existing CRL
Info: Using environment 'development'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.8.85)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent02
Info: Applying configuration version '1787123398'
Notice: /Stage[main]/Dynamicinfra::Install/Package[apache2]/ensure: created
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/lib/dynamicinfra]/ensure: cr
eated
Notice: /Stage[main]/Dynamicinfra::Config/File[/etc/dynamicinfra-development.con
f]/ensure: defined content as '{sha256}c00d6e52d05beef5c8169e632012d54ec7c64de90
b5c3e6765d72f61a9e38107'
Notice: /Stage[main]/Dynamicinfra::Config/File[/etc/apache2/sites-available/dyna
micinfra.conf]/ensure: defined content as '{sha256}217b187f24166a573ab44b3d1448b
535f6c800c12aa9d4d796c9169e087cd529'
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content
:
--- /var/www/html/index.html      2026-08-19 07:10:06.732594411 +0000

```

```

ubuntu@puppet-agent02: ~
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content
: content changed '{sha256}6faef4d5d777fdcaa653766b0ac8b9ed32d0fd87f7dcd79f02ff5
24dd1b0eb69' to '{sha256}6b23b822ff944502e84b35f8f9647f95c42d51e08b0fabd9b615ea0
3ee130803'
Info: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]: Scheduling
refresh of Service[apache2]
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'
from 1 event
Notice: /Stage[main]/Dynamicinfra::Report/File[/var/lib/dynamicinfra/system-repo
rt.txt]/ensure: defined content as '{sha256}c525c6c6c672444c9a62b55938fe8d823e62
314dda3683f4620c4c961569dfe7'
Notice: Applied catalog in 14.58 seconds
ubuntu@puppet-agent02:~$ sudo ls -l /etc/apache2/sites-available/dynamicinfra.co
nf
-rw-r--r-- 1 root root 454 Aug 19 07:10 /etc/apache2/sites-available/dynamicinfr
a.conf
ubuntu@puppet-agent02:~$ sudo ls -l /var/www/html/index.html
-rw-r--r-- 1 root root 653 Aug 19 07:10 /var/www/html/index.html
ubuntu@puppet-agent02:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset
: enabled)
   Active: active (running) since Wed 2026-08-19 07:10:14 UTC; 25min ago
     Docs: https://httpd.apache.org/docs/2.4/

```

```
ubuntu@puppet-agent02: ~  
Environment: development  
ubuntu@puppet-agent02:~$ sudo cat /etc/apache2/sites-available/dynamicinfra.conf  
<VirtualHost *:80>  
    ServerName WEB-02  
  
    DocumentRoot /var/www/html  
  
    <Directory /var/www/html>  
        Options Indexes FollowSymLinks  
        AllowOverride None  
        Require all granted  
    </Directory>  
  
    ErrorLog ${APACHE_LOG_DIR}/puppet-managed-error.log  
    CustomLog ${APACHE_LOG_DIR}/puppet-managed-access.log combined  
  
    Header always set X-Puppet-Environment "development"  
    Header always set X-Puppet-Role "web"  
</VirtualHost>  
ubuntu@puppet-agent02:~$ sudo cat /var/www/html/index.html  
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">
```

```
    <meta charset="UTF-8">  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
    <title>WEB-02</title>  
</head>  
<body>  
    <h1>WEB-02</h1>  
  
    <h2>Puppet Managed Web Server</h2>  
  
    <p><strong>Environment:</strong> development</p>  
    <p><strong>Server Role:</strong> web</p>  
  
    <h3>System Information</h3>  
  
    <p><strong>Hostname:</strong> puppet-agent02</p>  
    <p><strong>IP Address:</strong> 172.31.10.53</p>  
    <p><strong>Operating System:</strong> Ubuntu</p>  
    <p><strong>OS Family:</strong> Debian</p>  
  
    <hr>  
  
    <p>Managed automatically by Puppet.</p>  
</body>
```

```
ubuntu@puppet-agent02: ~  
<body>  
  <h1>WEB-02</h1>  
  
  <h2>Puppet Managed Web Server</h2>  
  
  <p><strong>Environment:</strong> development</p>  
  <p><strong>Server Role:</strong> web</p>  
  
  <h3>System Information</h3>  
  
  <p><strong>Hostname:</strong> puppet-agent02</p>  
  <p><strong>IP Address:</strong> 172.31.10.53</p>  
  <p><strong>Operating System:</strong> Ubuntu</p>  
  <p><strong>OS Family:</strong> Debian</p>  
  
  <hr>  
  
  <p>Managed automatically by Puppet.</p>  
</body>  
</html>  
ubuntu@puppet-agent02:~$ sudo cat /etc/dynamicinfra-development.conf  
This is the DEVELOPMENT configuration.  
ubuntu@puppet-agent02:~$ |
```

